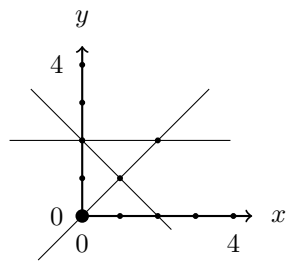
Linear Algebra 2024 Midterm Solutions

True or False Questions

Answer 'T(rue)' or 'F(alse)'. (You don't have to give a reason, but it might get you half a point if you are wrong.)

1. If columns 1 and 3 of B are the same, then columns 1 and 3 of AB are. T
2. $(AB)^2 = A^2B^2$ F
3. If A is invertible, then the left nullspace of A is isomorphic to the nullspace of A. T: A is square, so both are 0 dimensional subspaces of R^n
4. $((AB)^{-1})^T = (A^{-1})^T(B^{-1})^T$ T
5. Any three independent vectors in $\mathbb{R}^3$ span $\mathbb{R}^3$. T
6. A matrix M is invertible if and only if it has 3 pivots. F: Only true for 3×3 matrices.
7. Given a matrix A, if we reduce the augmented matrix $[A \mid I]$ by Gaussian Elimination to $[A' \mid B]$, then $BA = I$. F: $BA = A'$
8. The columns of a matrix are a basis of the column space. F: They CONTAIN a basis
9. The nullspace of the incidence matrix M of a graph G always has dimension 1. F: Only if G is connected.
10. A and A^T have the same rank. T
11. If $x \in W$ then $\text{proj}_W\, x = 0$. F: $\text{proj}_W\, x = x$
12. For a subspace S of $\mathbb{R}^n$ we have $(S^\perp)^\perp = S$. T: That $S \subset (S^\perp)^\perp$ is clear. Equality because $\dim(S) + \dim(S^\perp) = n$.
13. The projection matrix for the space generated by $(1, 0, 0)$ and $(1, 1, 0)$, is invertible. F: It has rank 2 so doesn't have full rank.

1. Given a system of three equations in two variables (x and y), where the equations define the shown lines, find all solutions to the system.



Solution

There are no solutions. There are no points in which all three lines intersect.

2. A system of linear equations has 3 equations in 4 variables. Describe what the set of solutions in $\mathbb{R}^4$ can be.

Solution

The set of solutions is the intersection of three (3-dimensional) hyperplanes in $\mathbb{R}^4$. The intersection can be a hyperplane (if the are all the same) a plane (if two are the same an one other is in general position to them) a line (if all three are in general position) or empty (if some two of them are parallel). $\boxed{\text{An affine space in } \mathbb{R}^4 \text{ of any dimension but 1 or 4.}}$ (I allowed 'any dimension but 1', because I guess all of the equations could have had all co-efficients 0.)

3. Find the solutions for the following system.

$$\begin{aligned} 2w + x &= 0 \\ w + 2x + y &= 0 \\ x + 2y + z &= 0 \\ y + 2z &= 5 \end{aligned}$$

Solution

Eliminating we get the system

$$\begin{array}{rcl} 2w + x & = 0 & \\ \frac{3}{2}x + y & = 0 & \\ \frac{4}{3}y + z & = 0 & \\ \frac{5}{4}z & = 5 & \end{array} \quad \rightsquigarrow \quad \begin{array}{rcl} w & = -1 & \\ x & = 2 & \\ y & = -3 & \\ z & = 4 & \end{array}$$

from which we get the solution $\boxed{\text{(w,x,w,z) = (-1,2,-3 ,4)}}$.

4. Find an LU decomposition of the matrix

$$A = \begin{bmatrix} 4 & 7 \\ 2 & 1 \end{bmatrix}.$$

Solution

Using Gauss Jordan elimination:

$$\left[\begin{array}{cc|cc} 4 & 7 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 4 & 7 & 1 & 0 \\ 0 & -\frac{5}{2} & -\frac{1}{2} & 1 \end{array}\right] =: [U \mid L^{-1}]$$

we have that $L^{-1}A = U$. As L^{-1} is the row matrix that takes half the first row from the second, so its inverse is the matrix that adds half the first row to the second:

$$L = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}.$$

So

$$\begin{bmatrix} 4 & 7 \\ 2 & 1 \end{bmatrix} = A = LU = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 4 & 7 \\ 0 & -\frac{5}{2} \end{bmatrix}$$

is an LU decomposition of A.

5. Which of the following subsets of $\mathbb{R}^3$ are subspaces (give a short reason for those that aren't):

   (a) The set of vectors (b_1, b_2, b_3) with $b_1 = 1$.

   (b) The set of vectors (b_1, b_2, b_3) with $b_1 = 0$.

   (c) The set of vectors (b_1, b_2, b_3) where b_1 is integer.

   (d) The set of vectors (b_1, b_2, b_3) with $b_1 b_2 = 0$.

   (e) The set of linear combinations of the vectors $(1, 2, 6)$ and $(3, 2, 1)$.

   (f) The set of vectors (b_1, b_2, b_3) with $b_1 + b_2 - 2b_3 = 3$.

Solution

These are: (b), (e).
(a) and (f) don't contain $(0,0,0)$ so are not closed under scalar multiplicaton.
(c) is not closed under scalar multiplication (by say $1/2$).
(d) is the union of two planes, (not the intersection): $(1,0,0)$ and $(0,1,1)$ are in the set, but their sum is not.

6. In the space M_3 of 3×3 real matrices, what is the smallest subspace that contains all symmetric matrices and all lower triangular matrices? What is the largest subspace contained in both of these subspaces?

Solution

The smallest subspace containing all symmetric and lower triangular matrices is $\boxed{M_3 \text{ itself}}$. To see this we express any matrix A as a sum of a symmetric matrix and a lower triangular matrix. Indeed, where $A = [a]_{i,j}$ let $L = [l]_{i,j}$ where $l_{ij} = a_{ij} - a_{ji}$ if $i \geq j$ and $l_{ij} = 0$ otherwise. Let $S = A - L$. By definition L is lower triangular, and S is symmetric. Both are in our space, and then so is $A = S + L$.
The largest space contained in the set of symmetric and lower triangular matrices is $\boxed{\text{the space of diagonal matrices}}$ in $\mathbb{R}^3$. Certainly this is the intersection of these sets, and it is a space; this is enough.

7. What is the *rank* of a matrix A?

Solution

Lots of ways one can describe this. Here are some:
- It is the number of pivots when we eliminate the matrix.
- The the dimension of the columnspace (or rowspace) of the matrix.
- For an $m \times n$ matrix A it is n minus the nullity of A.

8. The row-reduced echelon form of the matrix C is given below.

$$[C|I] = \left[\begin{array}{cccc|ccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 2 & 1 & 2 & 3 & 0 & 1 & 0 \\ 2 & 2 & 3 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow{elimination} \left[\begin{array}{cccc|ccc} 1 & 0 & 0 & 2 & 1 & 1 & -1 \\ 0 & 1 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 0 & -2 & 0 & 1 \end{array}\right]$$

(a) Find the solution of $Cx = 0$
(b) What are the dimensions of the nullspace and the left nullspace.
(c) Give a basis of the columnspace.

(d) Give a basis of the rowspace.

Solution

(a) As z is the only free variable solutions $\mathbf{x} = (w, x, y, z)^T$ are

$$\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = z \begin{bmatrix} -2 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

(b) The nullspace has dimension $\boxed{1}$, so the matrix has rank $r = 3$, so the left nullspace has dimension $m - r = 3 - 3 = \boxed{0}$.

(c) The columnspace has basis $\boxed{\{\ (1,2,2),\ (1,1,2),\ (1,2,3)\ \}}$. (Or any basis of $\mathbb{R}^3$ will do.)

(d) The rowspace has basis $\boxed{\{\ (1,0,0,2),\ (0,1,0,-1),\ (0,0,1,0)\}}$.

9. What multiple of $a = (2, 1, 1)$ is closest to the point $b = (2, 4, 4)$?

Solution

The projection of b onto a is $\frac{a^T b}{a^t a} a = \frac{12}{6} a = 2a$. So $\boxed{2a = (4,2,2)}$ is closest.

10. Suppose that P is the projection onto some subspace S of $\mathbb{R}^n$ and Q is the projection onto $S^\perp$. What are $P + Q$ and PQ. Show that $P - Q$ is its own inverse.

Solution

Let B_S be a basis of S and let B_Q a basis of Q. As they are orthogonal to each other and their size sums to n we have that $B = B_S \cup B_Q$ is a basis of $\mathbb{R}^n$.

Take any vector b in B. As b is either in B_S or B_Q we have that $(P+Q)b = Pb + Qb = b + 0$ or $0 + b$. So $(P+Q)b = b$. As this is true for all vectors in a basis of $\mathbb{R}^n$ it is true for all vectors, so $\boxed{\text{P+Q = I}}$.

$\boxed{\text{PQ} = 0}$ as Q projects any b onto the orthogonal complement of S, and so P projects it to 0. Or using the previous bit, $PQ = P(I - P) = P - P^2 = P - P = 0$ where $P^2 = P$ is because P is a projection, so idempotent.

$P - Q$ is its own inverse as $(P - Q)(P - Q) = P^2 - QP - PQ + Q^2$. We have that P and Q are projections, so idempotent, meaning that $P^2 = P$ and $Q^2 = Q$. Moreover, from above, $PQ = 0$, and by the same argument $QP = 0$, so

$$(P - Q)(P - Q) = P^2 - QP - PQ + Q^2 = P + Q = I,$$

as needed.